Newton's Laws

ClipSAT Printable Practice Suite (clipsat.org) • Topic 13.01

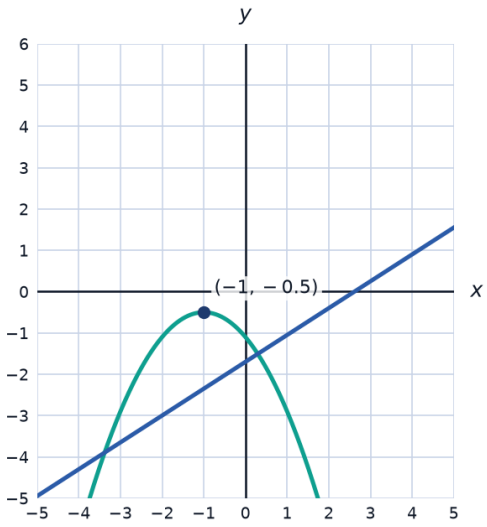
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Particles of mass 11 kg and 16 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Newton's Laws of Motion · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Express $\frac{3+\sqrt{5}}{2-\sqrt{5}}$ in the form $p + q\sqrt{5}$, where p and q are rational numbers to be found.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A circle has radius 8 cm. A sector of this circle has a total perimeter of 28 cm. Find the angle of the sector in radians, and calculate the exact area of the sector.

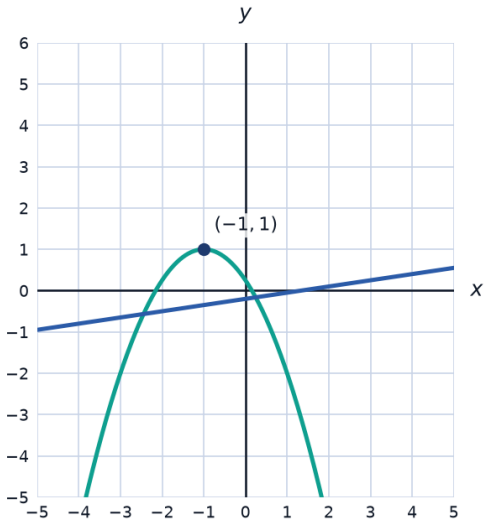
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the set of values of x satisfying the inequality $2x^2 - 7x + 3 \leq 0$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The curve $y = f(x)$ has a stationary point at $(4, -6)$. State the coordinates of the corresponding stationary point on the curve: (a) $y = f(x + 3) + 2$, (b) $y = 3f(2x)$, (c) $y = -f(-x)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A student computes the volume of a cylinder using $r = 3.2$ cm and $h = 7.85$ cm. The raw calculator display shows $252.518...$. How should the final answer be reported according to Cambridge AS Level guidelines?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A discrete random variable Y has $E(Y) = 5$ and $\text{Var}(Y) = 4$. Calculate $E(Y^2)$ and $\text{Var}(10 - 3Y)$.

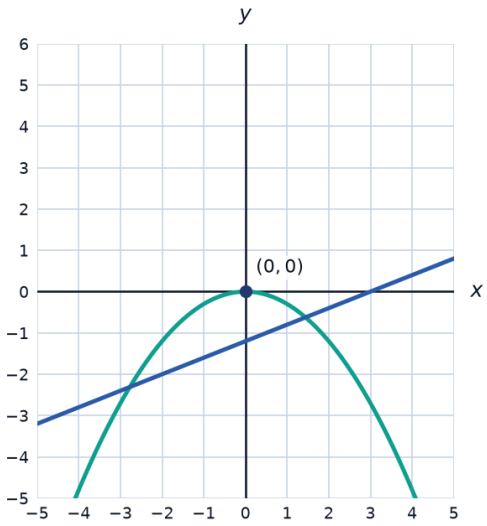
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A continuous dataset is grouped into class intervals. The interval $20 \leq x < 35$ has a frequency of 45. Find the frequency density for this class.

© ClipSAT — clipsat.org | All Rights Reserved

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Find the number of ways to arrange the 7 letters in the word SUCCESS.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Indices, Surds & Quadratics · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Differentiation & Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Simultaneous Equations & Inequalities · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Trigonometry · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

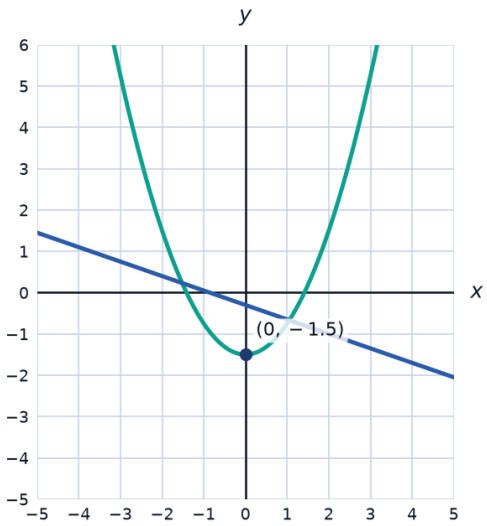
15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.01 Indices, Surds & Quadratics**

[About AS Level · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

© ClipSAT — clipsat.org | All Rights Reserved

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Series & Binomial Theorem**

[Coordinate Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Kinematics**

[Representation of Data · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 12.01 Forces & Equilibrium**

[Discrete Random Variables · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 14.01 Energy, Work & Power**

[Energy, Work & Power · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 15.01 Representation of Data**

[Functions: Composite & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A constant resultant force of 36 N acts on a mass of 4 kg. What is the acceleration produced?

a. $9\ \text{m/s}^2$

b. $144\ \text{m/s}^2$

c. $4\ \text{m/s}^2$

d. $6\ \text{m/s}^2$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A car of mass 1200 kg is traveling at 25 m/s when the brakes are applied, bringing it to rest in 50 m. Find the magnitude of the braking force.

a. 7500 N

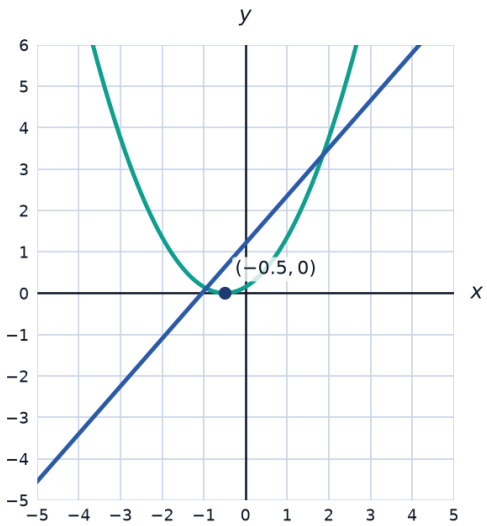
b. 6000 N

c. 9000 N

d. 15000 N

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An elevator of mass 800 kg accelerates upwards at $1.5\ \text{m/s}^2$. Taking $g = 10\ \text{m/s}^2$, find the tension in the cable.



a. 9200 N

b. 8000 N

c. 6800 N

d. 1200 N

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two masses of 3 kg and 2 kg are connected by a light inextensible string passing over a smooth fixed pulley (Atwood machine). Taking $g = 10\ \text{m/s}^2$, find the acceleration of the system.

a. $2\ \text{m/s}^2$

b. $1\ \text{m/s}^2$

c. $4\ \text{m/s}^2$

d. $5\ \text{m/s}^2$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the system in the previous question, find the tension T in the string.

a. $24\ \text{N}$

b. $25\ \text{N}$

c. $20\ \text{N}$

d. $30\ \text{N}$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the total force exerted by the string on the pulley in the previous system?

a. $48\ \text{N}$ downwards

b. $24\ \text{N}$ downwards

c. $50\ \text{N}$ downwards

d. $40\ \text{N}$ downwards

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A block of mass 5 kg is pulled up a smooth slope inclined at 30° to the horizontal by a force of 40 N acting parallel to the slope. Taking $g = 10\ \text{m/s}^2$, find the acceleration of the block.

a. $3\ \text{m/s}^2$

b. $5\ \text{m/s}^2$

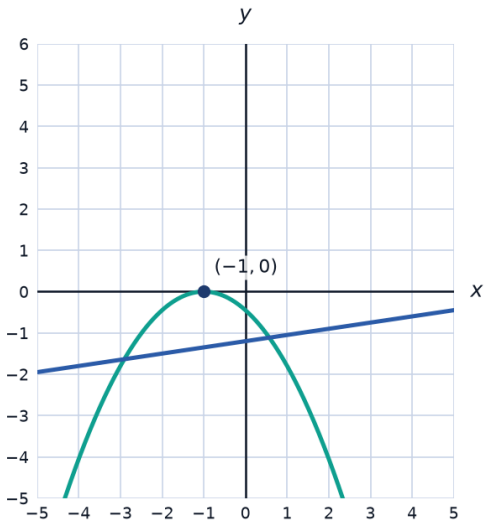
c. $2\ \text{m/s}^2$

d. $4\ \text{m/s}^2$

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If the slope in the previous question is rough with $\mu = 0.2$, what is the new acceleration of the block?



a. $1.27\ \text{m/s}^2$

b. $3.00\ \text{m/s}^2$

c. $2.15\ \text{m/s}^2$

d. $0.85\ \text{m/s}^2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A sled of mass 20 kg slides down an incline of 25° at constant speed. What is the coefficient of kinetic friction μ?

a. $\mu = \tan(25^\circ) \approx 0.466$

b. 0.423

c. 0.906

d. 0.500

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A block A of mass 4 kg rests on a smooth horizontal table and is connected by a light string over a smooth pulley at the edge of the table to a hanging block B of mass 1 kg. Taking $g = 10\ \text{m/s}^2$, find the acceleration.

a. $2\ \text{m/s}^2$

b. $2.5\ \text{m/s}^2$

c. $1.5\ \text{m/s}^2$

d. $10\ \text{m/s}^2$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If the table in the previous system has coefficient of friction $\mu = 0.15$, find the acceleration of the system.

a. $0.8\ \text{m/s}^2$

b. $1.2\ \text{m/s}^2$

c. $1.5\ \text{m/s}^2$

d. $0.5\ \text{m/s}^2$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A car of mass 1000 kg tows a trailer of mass 500 kg with acceleration $1.2\ \text{m/s}^2$. Resistance to motion is 300 N on the car and 200 N on the trailer. Find the tension in the towbar.

a. 800 N

b. 600 N

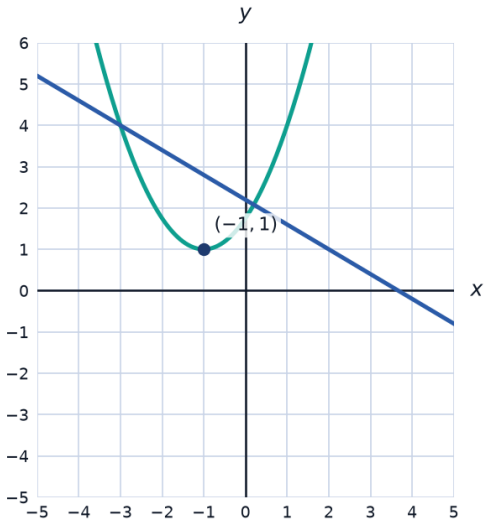
c. 1100 N

d. 1500 N

© ClipSAT — clipsat.org | All Rights Reserved

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A resultant force acts on a body of mass 8 kg, giving it an acceleration of 8 m/s^2. Find the magnitude of the resultant force.



a. 62

b. 64

c. 128

d. 66

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A body of mass 4 kg is acted on by a driving force of 13 N and a resisting force of 1 N. Find its acceleration.

a. 2

b. 13/4

c. 3

d. 4

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Balls are drawn one at a time, without replacement, from a bag of 20 balls; X is the number of red balls drawn. Is X binomially distributed?

a. Yes — but only if $n > 30$

b. Yes — fixed, independent trials with constant success probability

c. No — trials are not independent / probability is not constant

d. No — the number of trials is not fixed

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In how many distinct ways can 6 different people be seated around a circular table (rotations considered identical)?

a. 120

b. 103

c. 122

d. 100

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An engine exerts a driving force of 414 N on a vehicle moving at a constant 19 m/s. Find the power developed.

a. 7866

b. 7885

c. 15732

d. 7847

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 12.01 Forces & Equilibrium

Find the number of distinct arrangements of the letters of the word PUPPY.

a. 103

b. 27

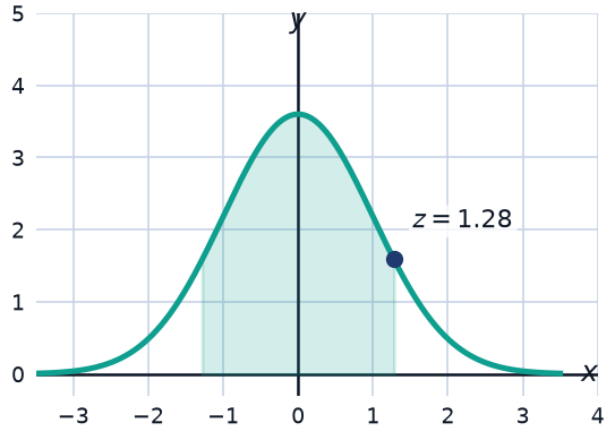
c. 64

d. 60

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 14.01 Energy, Work & Power

A random variable $X \sim B(100, 0.6)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.



a. 58.0

b. 65.0

c. 60

d. 62.0

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 15.01 Representation of Data

From a cumulative frequency curve with $n = 80$, the median is read at cumulative frequency:

a. 20

b. 40

c. 60

d. 80

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 16.01 Permutations & Combinations

Find the x-coordinates of the points of intersection of $y = x^2 - 9x + 22$ and $y = 2$.

a. x=4 or x=5

b. x=-4 or x=-5

c. x=5 or x=5

d. x=4 or x=4

© ClipSAT — clipsat.org | All Rights Reserved

42 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 17.01 Discrete Random Variables**

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-2;\ -6)$. State the coordinates of the new vertex.

a. (-2,6)

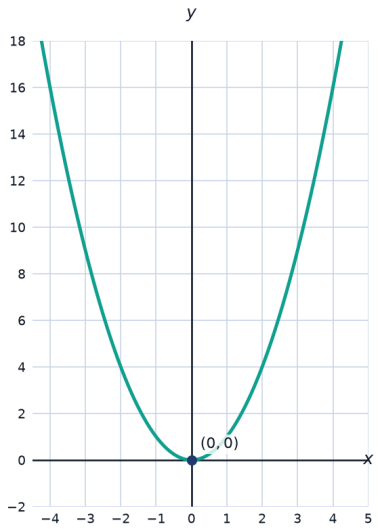
b. (-2,-6)

c. (-6,-2)

d. (2,-6)

43 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 18.01 Binomial Distribution**

The minimum point of $y = x^2$ is translated by the vector $(-3; 6)$ to give curve C. State the coordinates of the minimum point of C.



a. (3,6)

b. (6,-3)

c. (-3,6)

d. (-3,-6)

44 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 19.01 Normal Distribution**

The curve $y = x^2$ is translated by the vector $(7; 4)$. State the coordinates of the new vertex.

a. (-7,4)

b. (7,4)

c. (4,7)

d. (7,-4)

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the variance of the data set: $4, 8, 14, 9, 2, 6$.

a. 1/2

b. 1/4

c. 2/3

d. 533/36

© ClipSAT — clipsat.org | All Rights Reserved